WEEK 9 – SALES AND CUSTOMER ANALYTICS SYSTEM

Project Title: Automobile Accessories and Spare Parts Shopping Platform

Subject: Database Management Systems (DBMS)

1. Introduction

Auto Hub Shop is an automobile accessories and spare parts shopping platform developed using a Database Management System. The project manages customer details, product information, orders, and sales transactions.

Week 9 focuses on the Sales and Customer Analytics System. SQL aggregate functions such as `COUNT()`, `SUM()`, `AVG()`, `MIN()`, and `MAX()` are used to analyze sales data and generate business reports. The system helps identify customer purchasing patterns, total revenue, order values, and customer spending.

2. Scenario

The management team of Auto Hub Shop wants to analyze its sales data to understand customer purchasing behaviour, product performance, and revenue trends.

To achieve this, SQL queries are used to calculate the total number of orders, total revenue, average order value, and highest and lowest order amounts. Customer purchase analysis is performed to identify top customers, frequent customers, total spending, and average spending. The system also includes product performance analysis and category-wise sales analysis to identify best-selling products and revenue-generating categories.

The generated reports help management monitor business performance, understand customer preferences, and make informed decisions regarding sales, inventory, and promotional activities.

3. Sales and Customer Analytics Reports

3.1 Apply SQL Aggregate Functions

3.1.1 Total Number of Orders

3.1.2 Total Revenue Generated

3.1.3 Average Order Value

3.1.4 Highest and Lowest Order Amount

3.1.5 Total Sales for a Specific Period

```
SQL> SELECT COUNT(*) AS total_orders FROM orders;

TOTAL_ORDERS
------------
          10

SQL> SELECT SUM(total_amount) AS total_revenue FROM orders;

TOTAL_REVENUE
-------------
      1129000

SQL> SELECT AVG(total_amount) AS average_order_value FROM orders;

AVERAGE_ORDER_VALUE
-------------------
             112900

SQL> SELECT MAX(total_amount) AS highest_order, MIN(total_amount) AS lowest_order FROM orders;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
       300000         9000

SQL> SELECT SUM(total_amount) AS period_sales FROM orders WHERE order_date >= DATE '2026-09-01' AND order_date < DATE '2026-10-01';

PERIOD_SALES
------------
      675000
```

3.2 Customer Purchase Analysis

3.2.1 Total Number of Orders by Each Customer

```
SQL> SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders FROM customer c LEFT JOIN orders o ON c.customer_id = o.customer_id GR
OUP BY c.customer_id, c.customer_name;

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_ORDERS
-------------------- -------------------- ------------
C-2001               Lira                            2
C-2002               Kenny                           2
C-2003               Kairo                           2
C-2004               Suxi                            2
C-2005               Blissy                          2
C-2006               Nova                            0
C-2010               Ash                             0
C-2007               Kaito                           0
C-2008               Valtor                          0
C-2009               Infernape                       0

10 rows selected.
```

3.2.2 Total Amount Spent by Each Customer

```
SQL> SELECT c.customer_id, c.customer_name, NVL(SUM(o.total_amount), 0) AS total_spent FROM customer c LEFT JOIN orders o ON c.customer_id = o.custo
mer_id GROUP BY c.customer_id, c.customer_name;

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_SPENT
-------------------- -------------------- -----------
C-2001               Lira                       54000
C-2002               Kenny                     155000
C-2003               Kairo                     220000
C-2004               Suxi                      355000
C-2005               Blissy                    345000
C-2006               Nova                           0
C-2010               Ash                            0
C-2007               Kaito                          0
C-2008               Valtor                         0
C-2009               Infernape                      0

10 rows selected.
```

3.2.3 Average Spending per Customer

```
SQL> SELECT c.customer_id, c.customer_name, NVL(AVG(o.total_amount), 0) AS average_spending FROM customer c LEFT JOIN orders o ON c.customer_id = o.
customer_id GROUP BY c.customer_id, c.customer_name;

CUSTOMER_ID          CUSTOMER_NAME        AVERAGE_SPENDING
-------------------- -------------------- ----------------
C-2001               Lira                          27000
C-2002               Kenny                         77500
C-2003               Kairo                        110000
C-2004               Suxi                         177500
C-2005               Blissy                       172500
C-2006               Nova                              0
C-2010               Ash                               0
C-2007               Kaito                             0
C-2008               Valtor                            0
C-2009               Infernape                         0

10 rows selected.
```

3.2.4 Customers with Maximum Purchase Amount

```
SQL> SELECT c.customer_id, c.customer_name, SUM(o.total_amount) AS total_spent FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP
BY c.customer_id, c.customer_name HAVING SUM(o.total_amount) = (SELECT MAX(customer_total) FROM (SELECT SUM(total_amount) AS customer_total FROM ord
ers GROUP BY customer_id));

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_SPENT
-------------------- -------------------- -----------
C-2004               Suxi                      355000
```

3.2.5 Customers with Fewer Purchases

```
SQL> SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders FROM customer c LEFT JOIN orders o ON c.customer_id = o.customer_id GR
OUP BY c.customer_id, c.customer_name HAVING COUNT(o.order_id) <= 1;

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_ORDERS
-------------------- -------------------- ------------
C-2006               Nova                           0
C-2010               Ash                            0
C-2007               Kaito                          0
C-2008               Valtor                         0
C-2009               Infernape                      0
```

3.3 Top Customers Based on Purchase Amount

3.3.1 Top Five Customers by Total Spending

3.3.2 Customers with Maximum Number of Orders

3.3.3 Frequent Customers

3.3.4 High-Value Customers

```
SQL> SELECT * FROM (SELECT c.customer_id, c.customer_name, SUM(o.total_amount) AS total_spent FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY
c.customer_id, c.customer_name ORDER BY total_spent DESC) WHERE ROWNUM <= 5;

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_SPENT
-------------------- -------------------- -----------
C-2004               Suxi                      355000
C-2005               Blissy                    345000
C-2003               Kairo                     220000
C-2002               Kenny                     155000
C-2001               Lira                       54000

SQL> SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id, c
.customer_name HAVING COUNT(o.order_id) = (SELECT MAX(order_count) FROM (SELECT COUNT(*) AS order_count FROM orders GROUP BY customer_id));

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_ORDERS
-------------------- -------------------- ------------
C-2001               Lira                            2
C-2002               Kenny                           2
C-2003               Kairo                           2
C-2004               Suxi                            2
C-2005               Blissy                          2

SQL> SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id, c
.customer_name HAVING COUNT(o.order_id) >= 2;

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_ORDERS
-------------------- -------------------- ------------
C-2001               Lira                            2
C-2002               Kenny                           2
C-2003               Kairo                           2
C-2004               Suxi                            2
C-2005               Blissy                          2

SQL> SELECT c.customer_id, c.customer_name, SUM(o.total_amount) AS total_spent FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id,
c.customer_name HAVING SUM(o.total_amount) >= 50000 ORDER BY total_spent DESC;

CUSTOMER_ID          CUSTOMER_NAME        TOTAL_SPENT
-------------------- -------------------- -----------
C-2004               Suxi                      355000
C-2005               Blissy                    345000
C-2003               Kairo                     220000
C-2002               Kenny                     155000
C-2001               Lira                       54000
```

3.4 Best-Selling Product Analysis

3.4.1 Products with Maximum Sales Quantity

```
SQL> SELECT p.product_id, p.product_name, SUM(od.quantity) AS quantity_sold FROM product p JOIN order_details od ON p.product_id = od.product_id GROUP BY p.product_id
, p.product_name HAVING SUM(od.quantity) = (SELECT MAX(total_quantity) FROM (SELECT SUM(quantity) AS total_quantity FROM order_details GROUP BY product_id));

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
P2002
Castrol Engine Oil
          25
```

3.4.2 Products Generating the Highest Revenue

```
SQL> SELECT p.product_id, p.product_name, SUM(od.quantity * od.price) AS revenue FROM product p JOIN order_details od ON p.product_id = od.product_id GROUP BY p.produ
ct_id, p.product_name ORDER BY revenue DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
   REVENUE
----------
P2001
Brake Pad Set
    427500

P2003
Car Phone Holder
    336000

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
   REVENUE
----------

P2002
Castrol Engine Oil
     87475

P2004
Bike Crash Guard

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
   REVENUE
----------
     50000
```

3.4.3 Least-Selling Products

```
SQL> SELECT p.product_id, p.product_name, NVL(SUM(od.quantity), 0) AS quantity_sold FROM product p LEFT JOIN order_details od ON p.product_id = od.product_id GROUP BY
 p.product_id, p.product_name ORDER BY quantity_sold ASC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
P2011
Mobile Holder
            0

P2007
Leather Seat Cover
            5

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------

P2009
LED Headlight
            6

P2003
Car Phone Holder

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
            8

P2010
New Car Perfume
           10
```

3.4.4 Products Requiring Promotion

```
SQL> SELECT p.product_id, p.product_name, NVL(SUM(od.quantity), 0) AS quantity_sold FROM product p LEFT JOIN order_details od ON p.product_id = od.product_id GROUP BY
 p.product_id, p.product_name HAVING NVL(SUM(od.quantity), 0) <= 1;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
P2011
Mobile Holder
            0
```

3.5 Category-Wise Sales Analysis

3.5.1 Total Sales Generated by Each Category

```
SQL>  SELECT c.category_id, c.category_name, NVL(SUM(od.quantity * od.price), 0) AS total_sales FROM category c LEFT JOIN product p ON c.category_id = p.category_id L
EFT JOIN order_details od ON p.product_id = od.product_id GROUP BY c.category_id, c.category_name;

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
Cat0014
Bike Accessories
      50000

Cat0015
Helmets
      42000

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------

Cat0016
Batteries
      36000

Cat0017
Seat Covers

CATEGOR
-------
CATEGORY_NAME
```

3.5.2 Number of Products Sold in Each Category

```
SQL>  SELECT c.category_name, NVL(SUM(od.quantity), 0) AS products_sold FROM category c LEFT JOIN product p ON c.category_id = p.category_id LEFT JO
IN order_details od ON p.product_id = od.product_id GROUP BY c.category_name;

CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Bike Accessories
           10

Helmets
           12

Batteries
           20


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Seat Covers
            5

Floor Mats
           15

LED Lights
            6


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
```

3.5.3 Highest Revenue-Generating Category

```
SQL>  SELECT * FROM (SELECT c.category_id, c.category_name, NVL(SUM(od.quantity * od.price), 0) AS total_revenue FROM category c LEFT JOIN product p
 ON c.category_id = p.category_id LEFT JOIN order_details od ON p.product_id = od.product_id GROUP BY c.category_id, c.category_name ORDER BY total_
revenue DESC) WHERE ROWNUM = 1;

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Cat0011
Spare Parts
       427500
```

3.5.4 Average Sales per Category

```
SQL>  SELECT c.category_name, NVL(AVG(od.quantity * od.price), 0) AS average_sales FROM category c LEFT JOIN product p ON c.category_id = p.category
_id LEFT JOIN order_details od ON p.product_id = od.product_id GROUP BY c.category_name;

CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Bike Accessories
        50000

Helmets
        42000

Batteries
        36000


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Seat Covers
        32500

Floor Mats
        18000

LED Lights
        45000


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
             10     1129000             112900             300000
```

3.6.2 Customer Analytics Report

```
SQL> SELECT c.customer_name, COUNT(o.order_id) AS number_of_orders, NVL(SUM(o.total_amount), 0) AS total_spending FROM customer c LEFT JOIN orders o
 ON c.customer_id = o.customer_id GROUP BY c.customer_id, c.customer_name;

CUSTOMER_NAME        NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------- ---------------- --------------
Lira                                2          54000
Kenny                               2         155000
Kairo                               2         220000
Suxi                                2         355000
Blissy                              2         345000
Nova                                0              0
Ash                                 0              0
Kaito                               0              0
Valtor                              0              0
Infernape                           0              0

10 rows selected.
```

3.6.3 Product Performance Report

```
SQL> SELECT p.product_name, NVL(SUM(od.quantity), 0) AS quantity_sold, NVL(SUM(od.quantity * od.price), 0) AS revenue_generated FROM product p LEFT
JOIN order_details od ON p.product_id = od.product_id GROUP BY p.product_id, p.product_name ORDER BY revenue_generated DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Brake Pad Set
           15            427500

Car Phone Holder
            8            336000

Castrol Engine Oil
           25             87475


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Bike Crash Guard
           10             50000

LED Headlight
            6             45000

Steel Helmet
           12             42000


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Exide Car Battery
           20             36000
```

3.6.4 Category Analysis Report

```
SQL> SELECT c.category_name, NVL(SUM(od.quantity), 0) AS total_products_sold, NVL(SUM(od.quantity * od.price), 0) AS total_revenue FROM category c LEFT JOIN product p ON c.category_id = p.category_id LEFT JOIN order_detail
s od ON p.product_id = od.product_id GROUP BY c.category_id, c.category_name ORDER BY total_revenue DESC;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Spare Parts
                 15        427500

Car Accessories
                  8        336000

Engine Oils
                 25         87475


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Bike Accessories
                 10         50000

LED Lights
                  6         45000

Helmets
                 12         42000


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Batteries
                 20         36000

Seat Covers
                  5         32500

Car Perfumes
                 10         22000


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Floor Mats
                 15         18000

Mobile Holders
                  0             0
```

4. Result

The Week 9 Sales and Customer Analytics System was implemented using SQL queries in the Auto Hub Shop database. The results show that the database contains 10 orders, with total revenue of 1,129,000 and an average order value of 112,900. The highest order

amount is 300,000, while the lowest is 9,000. Total sales for September 2026 amount to 675,000.

Customer purchase analysis identifies Suxi as the highest-spending customer, with 355,000, followed by Blissy with 345,000. The reports demonstrate the use of SQL aggregate functions and customer analysis queries to summarize business data. The remaining product and category analysis results can be documented using their corresponding query output screenshots.